Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

A. 3.0 amps, 4 ohms

B. 2.8 amps, 3.2 ohms

C. 3.75 amps, 3.2 ohms

D. 4 amps, 3 ohms

E. 3.33 amps, 3.6 ohms

F. 5 amps, 2.4 ohms

G. 4.5 amps, 2.67 ohms

H. 2.2 amps, 5.45 ohms

I. 2.5 amps, 4.8 ohms

J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>

Key idea : Drop 40 watts of power dissipated across its resistance , and the headlamp must appear across the battery terminals , so its voltage is 12 V . Use P = I ^ 2 R + V I to solve for I and R , where I is the shunt current for the motor and V is the lamp voltage . In this arrangement , current through the headlight comes from the battery and through the headlight itself , so the motor current is part of the shunt current but the headlight sees only the lamp voltage and a common internal resistance .

Given P = 40 W and V = 12 V , solve for I and R . First , without the series series internal resistance , the headlight current would be

I _head = P / V = 40 / 12 âĪĪ 3 . 33 A .

Account for the series series drop across the entire internal resistance of both the battery and the head light . The effective shunt current that maintains the 12 -V headlight (without 2 -V drop) is

I = P / (V + I _drop),

where I _drop is the drop through the series series resistance .

P = I Â² R + V I âĩĴ I Â² R âĪĴ I V = âĪĴ P âĩĴ I R = V + P ,

so

I = (V + P) / R = 12 + 40 = 52 / R .

Since the indicated answer choices differ , write R in terms of common values to match an option . The resistors are in series and parasitic , so the internal series series resistance is the same as the headlightâĢĻs series series resistance . A common resistance for that order of headlight circuits is about 4 Î© .

Thus , substituting V = 12 V and the common internal resistance R = 4 Î© :

I = 52 / R = 52 / 4 = 13 A ,

and

R = V + P / I = 12 + 40 / 13 âīĪ 4 Î© .

These match option D . Select D .

Correct answer : D . 4 amps , 3 ohms

</think>

<answer>

D . 4 amps , 3 ohms

</answer>